

Sample Examination

Productions & Logistics

Summer Semester 2020 – Heilbronn Campus

July xx, 2020, Heilbronn / Online

Round numbers to the second decimal (e.g., 0.455 to 0.46 or 0.454 to 0.45) unless otherwise stated.

Please note that the language in this exam is not indicative of answers, and (potential) grammar errors do not render an otherwise correct true statement into a false one.

Available time: 120 minutes.

Module: Production

Exercise 1: Multiple Choice Questions (12 Points)

- 1.1** What of the following statements best describes process capacity?
- a) The average amount a process produces in a given amount of time.
 - b) The amount of time it requires to produce a given number of units.
 - c) The maximum amount that a process can produce in a given amount of time.
 - d) The maximum number of units a process can produce.
- 1.2** Which of the following statements relates to the fundamental tradeoff of capacity management?
- a) If demand is fluctuating, you cannot have 100% service level and 100% capacity utilization (without inventory buffers).
 - b) Capacity cannot be planned long-term and short-term.
 - c) Whether to manage capacity in services or production.
 - d) Whether to buy or lease assets.
- 1.3** Assume all EOQ model assumptions apply for a product (i.e., infinite replenishment rate). Assume you consider producing the product (i.e., finite production rate) incurring the same per-setup costs and the same holding costs and the same demand rate. Which of the statements is, in general, correct?
- a) Producing the product in-house, you will have the same batch sizes.
 - b) Producing the product in-house, you will have smaller batch sizes.
 - c) Producing the product in-house, you will have larger batch sizes.
 - d) Producing the product in-house is always more expensive.
- 1.4** Consider a product with constant demand rate of 150 products per year, annual holding costs of 24€ per product and per-setup costs of 200€ and production rate of 200 products per year. What is the optimal batch size?
- a) 50
 - b) 75
 - c) 100
 - d) None of the above
- 1.5** Consider a product with constant demand rate of 150 products per year, annual holding costs of 24€ per product and per-setup costs of 200€ and production rate of 200 products per year. How many times should you start producing a new batch per year on average?
- a) 3
 - b) 1.5
 - c) $2/3 = 0.66666$
 - d) None of the above

- 1.6** Consider the following description: “*** is a tool to create a better understanding of relationships between processes within a firm; it can help to identify bottlenecks”. Which concept is best described by this statement (i.e., what is ***)?
- a) A time series
 - b) A process flow diagram
 - c) The critical path method
 - d) Parallel process
- 1.7** Consider the following description: “*** is the task with the smallest capacity in a process”. Which concept is best described by this statement (i.e., what is ***)?
- a) The bottleneck
 - b) A sequential task
 - c) A parallel task
 - d) Demand
- 1.8** Consider the following description: “The *** is typically reached after a ramp-up phase”. Which concept is best described by this statement (i.e., what is ***)?
- a) throughput time
 - b) cycle time
 - c) steady state
 - d) downtime
- 1.9** Consider the following description: “*** is a tool to understand how different resources may overlap taking a stronger focus on timing”. Which concept is best described by this statement (i.e., what is ***)?
- a) A Gantt chart
 - b) A process flow diagram
 - c) The critical path method
 - d) A parallel process
- 1.10** Which of the following situations describes best a situation where a simulation is surely needed?
- a) When systems are too complex and there is too much uncertainty involved.
 - b) When a simple system is new and has not yet been implemented.
 - c) When there are many (e.g., more than 20) sequential tasks involved and one seeks to find the bottleneck.
 - d) If you want to do Kaizen.

1.11 Consider the four production systems called (1) make-to-order, (2) assemble-to-order, (3) make-to-stock, and (4) engineer-to-order. These are associated with different levels of inventory investment. Sort these four in increasing order, that is, start with the one that requires the lowest inventory investment. Which of the following statements appropriately matches the correct sequence?

- a) 1, 2, 3, 4
- b) 4, 3, 2, 1
- c) 4, 1, 2, 3
- d) 3, 2, 1, 4

1.12 Which of the following situations sounds like a legit recommendation to improve production (at least something you should always investigate)?

- a) Transform a pull production into push production.
- b) Transform a push production into a pull production.
- c) Increase muda.
- d) Minimize the return on net assets.

Exercise 2: Forecasting (12 points)

Consider the following time series.

Period	Actual demand
1	135
2	215
3	175
4	214
5	159

- a) Assume you are in period one and forecast period 2. Then assume you are in period 2 and forecast period 3. Continue until you forecast period 5. Use the following forecasting methods to make these forecasts (6 points):
- Naïve
 - Naïve with trend
 - Exponential smoothing with $\alpha = 0.6$, $F_0 = 100$
- b) For the periods 3-5, calculate the MAD and MSE of each method. Which method performs best and why? (4 points)
- c) Assume α was not given. Explain briefly how you could use Excel to find α . (2 points)

Exercise 3: Process Analysis (12 points)

- a) Critically discuss the following statement: "Throughput time is always strictly larger than customer order lead time."
(3 points)

Consider the hairdresser firm *ProfiCutZ*. Customers of this hairdresser proceed according to the following steps. They enter the shop immediately check in, which takes 2 minutes per customer. Then customers wait 7 minutes. Then they proceed to hair-washing, which takes 6 minutes per customer. Then they wait again, for 3 minutes. Next their hair is being cut by a professional hairdresser (this step is called hairdressing), which takes 30 minutes. After waiting another 5 minutes, customers proceed to check-out to pay, and leave the shop. This entire check-out procedure takes another 3 minutes.

Assume there are three groups of workers in the firm *ProfiCutZ*: professional hairdressers, unskilled hair washers, and administration. Professional hairdressers can perform any task, but usually only engage in hairdressing. Unskilled hair washers can only wash hair. Administration staff can only accommodate the check-in and check-out procedures.

Currently, *ProfiCutZ* employs 5 hairdressers, 2 unskilled hair washers, and 1 administrative person.

Perform a process flow chart analysis for the firm ProfiCutZ and answer the following questions.

- b) What is the capacity of ProfiCutZ in customers per hour (2 points)?
c) Which resource is the bottleneck at ProfiCutZ (1 point)?
d) How many customers are there, on average, at the same time at ProfiCutZ in the steady state (1 points)?
e) How many customers can be served per hour at ProfiCutZ if two further hairdressers were employed? Explain. (5 points)

Exercise 4: EOQ and lean (12 points)

- a) Assume your firm operates a number of $n \geq 2$ warehouses. Each has the same, stable demand of λ , annual per-unit holding costs of h , and per-order shipment costs of K . There is an offer on the table to pool all demand and operate only one warehouse to serve all customers. Assuming that this comes without any further cost increases, what percentage of your logistics costs (=holding costs + per setup costs) would your firm save? (4 points)
- b) Consider a continuous product (e.g. a liquid or gas). Assume per-setup costs of \$12, per-unit annual holding costs \$2, constant and known demand rate 57 units per week. Assume that your selling season lasts for 12 weeks. How often should you order during these 12 weeks? (5 points)
- c) Think about a product that is usually made to stock and name this product. Explain in 1 sentence why this product is made to stock. (1 point)
- d) Briefly describe at least two different steps that would be required to transform the push production of this product into a pull production. (2 points)

Exercise 5: Practice Problems (12 points)

- a) Consider Kristen Cookies Company as described in the case that was discussed in class. Assume this case was to take place today. Name two technologies that Kristen Cookies Company could use (in today's world). Which operations related KPIs that we discussed in class would be affected by these technologies and how? (6 points).

For the remainder of this task, consider the case OceanCove restaurant as we discussed it in class. Assume this restaurant has the exact same processes but in today's time. As manager of this restaurant you are confronted with the following offer: "You can use FoodX, a leading food delivery service and online platform provider, to efficiently deliver take-out menus to customers who are not dining in." (Hint: since the case and the solution from the class is available, you may want to refer to the numbers/data in the case.)

- b) State and explain the two most important benefits from an operations perspective related to the proposal (2 points).
- c) State and explain the two most important downsides from an operations perspective related to the proposal (2 points).
- d) Would you recommend using FoodX? Provide a quantitative reasoning (2 points)?

Module: Logistics

Exercise 1: Multiple Choice Questions (12 Points)

- 1.1** Consider the following term “strategic forward buying in sales promotions” which is described in an online lecture. Which of the following statements is correct?
- e) This relates to a mitigation strategy against the bullwhip effect.
 - f) This was not mentioned in either of the videos related to the bullwhip effect (and does not play a role).
 - g) This relates to a cause underlying the bullwhip effect.
 - h) This relates to a negative consequence of the bullwhip effect.
- 1.2** Consider the following term “Every Day Low Pricing (EDLP)” which is described in an online lecture. Which of the following statements is correct?
- a) This relates to a negative consequence of the bullwhip effect.
 - b) This was not mentioned in either of the videos related to the bullwhip effect (and does not play a role).
 - c) This relates to a mitigation strategy against the bullwhip effect.
 - d) This relates to a cause underlying the bullwhip effect.
- 1.3** Which distribution network design is similar to drop-shipping, except that the orders coming from different locations are combined so that the customer gets a single delivery?
- a) Manufacturer storage with direct shipping
 - b) Manufacturer storage with direct shipping and in-transit merge
 - c) Distributor storage with package carrier delivery
 - d) Distributor storage with manufacturer pickup
- 1.4** Which distribution network design is being used when inventory is not held by manufacturers at the factories, but is held by distributors/retailers in intermediate warehouses and package carriers are used to transport products from the intermediate location to the final customer?
- a) Distributor storage with package carrier delivery
 - b) Distributor storage with last-mile delivery
 - c) Manufacturer storage with direct shipping
 - d) Distributor storage with manufacturer pickup
- 1.5** Which of the following represents the “changing cells” in Excel Solver?
- a) Decision variables
 - b) Objective function
 - c) Constraint LHS

d) Non-negativity

- 1.6** The Hotelling model indicates that, under certain assumptions, firms will be located in close proximity to their competitors. Which of the following statements best describes the intuition about Hotelling competition?
- a) This model captures the behavior of hotels and they tend to be always close to touristic places.
 - b) Firms seek to be “at the center” so that they get the largest share of customers. If a firm were off center, another firm could move close to capture the larger share of customers.
 - c) Firms are in proximity to competitors because of governmental intervention.
 - d) The Hotelling Directive (93/13/EEC) prescribes firms with more than 1 billion revenue in the European Union (EU) not to be more than 5 km away from their competitors to ensure for perfect competition, to reduce consumer prices and thus inflation.
- 1.7** Which of the following correctly describes the complement of event A ?
- a) The complement of event A is the probability that event A does not occur.
 - b) The complement of event A is the set of all outcomes in event A .
 - c) The complement of event A is the set of outcomes which are in the sample space but not in A .
 - d) The complement of event A is the set of all outcomes in the sample space.
- 1.8** Which of the following statement is **incorrect** when comparing normal and Poisson distributions?
- a) The Poisson distribution depends on one parameter λ and the normal distribution depends on two parameters (μ, σ) .
 - b) The mean of both distributions is equal to their variances, respectively.
 - c) Normal distribution is always symmetric and Poisson distribution is always non-symmetric.
 - d) Poisson distribution is discrete while a normal distribution is continuous.
- 1.9** Which of the following comparisons between the order-up-to model and the newsvendor model is **true**?
- a) Both models sell leftover inventory at a discounted price.
 - b) The decision maker chooses the order-up-to level in the order-up-to model and the order quantity in the newsvendor model.
 - c) The newsvendor model should be applied when demand is uncertain and the order-up-to model should be applied when demand is constant.
 - d) In general, in both models it can be shown that the service level of 100% is optimal for the decision maker.

1.10 Consider the following scenario: There are 50 boxes of cereal in inventory before demand occurs. Demand in one period is 80 boxes. This means at the end of the period, on-hand inventory is A boxes and the backorder is B boxes.

- a) $A = 50, B = 30$
- b) $A = 50, B = 0$
- c) $A = 0, B = 30$
- d) $A = 50, B = 80$

1.11 Which of the following is the most likely correct statement?

- a) Due to the Corona virus crisis, there will be more global sourcing.
- b) Due to the Corona virus crisis, European pharmaceutical firms will offshore their critical production facilities to the U.S.
- c) The Corona virus crisis and the associated shutdown of only a few months was too short to have lasting impact on supply chain design, which tends to be a long-term decision.
- d) The Corona crisis heavily impacts most (global) supply chains and is likely to lead to changes in supply chain designs and distribution design networks.

1.12 Which of the following is the most likely correct statement?

- a) The Corona crisis leads to reduced consumer demand and therefore less bullwhip effect impact.
- b) The bullwhip effect is not a large issue anymore (due to digital transformation of most businesses) and thus is not related to the Corona virus crisis.
- c) The Corona virus crisis has led to more volatile demand (spiking for some products, dropping for others) while also leading to supply shocks. Taken together, both can have drastic consequences that can be exaggerated by the bullwhip effect.
- d) The bullwhip effect has been there before Corona and has nothing to do with it.

Exercise 2: Inventory Management (12 points)

You are operating a small book shop in Heilbronn and you are selling the book “*How to fail*”. You can place a replenishment order once per week. There is a lead time of two weeks. Space is expensive. You incur holding costs of 30% of the books’ value per year. If you run into a stock-out, items are back-ordered, that is, there are no lost sales. Yet you incur opportunity costs (these customers are unhappy); per unit back ordered per week you incur costs of 15% of the items value. Assume demand in each week follows a Poisson distribution with $\lambda = 3$.

(Hint: remember, you are allowed to use Excel but need to briefly explain what you did or add a screenshot!)

- a) Calculate the optimal service level. (1.5 points)
- b) What is the optimal in-stock probability? (0.5 points)
- c) If you choose a service level of 95%, what is the corresponding order-up-to level? (3 points)
- d) Consider an order-up-to level of 15 books. How likely will you face a stock-out? (2 point)
- e) Now assume you are in charge of managing the distributor’s inventory. You consider an order up-to policy. Per unit you incur annual holding costs of 35% of the items value, your back-order costs are 20% of the items value. Your order lead time is 8 weeks. Each week has normally distributed demand with $\mu = 300$ and $\sigma = 120$. What is your optimal order-up-to level? (3 points)
- f) Comment on the following statement: “A random variable is a function and can be seen as a model of reality.” (2 points)

Exercise 3: Supply Chain Coordination (12 points)

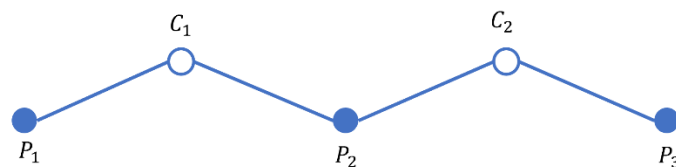
Assume that you use an order-up-to inventory policy. Your firm has signed a service level agreement on 98%. **That is, you must ensure to have a service level of at least 98%.**

- a) Demand is normally distributed, $\mu = 1000$, $\sigma = 30$. What is the optimal order-up-to level assuming that you may order a non-integer amount? (2 points)
- b) Using optimal decisions in the order-up-to model and the service level agreement as stated above, what is the standard deviation of your orders? Explain. (4 points)
- c) Imagine you are batching orders with a batch size of 25 units. Will the standard deviation of your orders remain, increase, or decrease? Why? Explain. (5 points)
- d) How is the effect in (c) be referred to in literature (and our lecture)? (1 point)

Exercise 4: Supply Chain Network Design (12 points)

Assume your firm has two customers (C_1 and C_2) and must serve both. You can decide to open up any number of three potential plants (P_1, P_2, P_3). But not each potential location can serve each customer. The costs associated with opening these plants are 6, 2, and 3, respectively. The following graph indicates which plants can serve which customer (indicated by straight lines; for instance, P_1 can serve C_1 ; but P_2 cannot serve C_1 .)

- Production plant
- Customer location



- a) Formulate this problem as a mixed integer linear program and introduce the decision variables. (1.5 points)
- b) Solve this problem applying the location covering problem heuristic as discussed in class. (2 points)
- c) Does the heuristic, if correctly applied, find the optimal solution in this case? Provide a proof. (1.5 points)
- d) Forget the problem above and think of the symmetric travelling salesmen problem (TSP). The subtour elimination constraint as introduced in the class states:

$$\sum_{(i,j) \in E: i \in S, j \notin S} x_{ij} + \sum_{(j,i) \in E: i \in S, j \notin S} x_{ji} \geq 2, \quad S \subset V, 2 \leq |S| \leq \left\lfloor \frac{|V|}{2} \right\rfloor$$

Why is it sufficient to say $|S| \leq \left\lfloor \frac{|V|}{2} \right\rfloor$? Provide a formal argument/proof. (4 points)

- e) Comparing the symmetric traveling salesmen problem (TSP) with shortest path problems between two cities, we noted that the latter can be solved efficiently using Dijkstra's Algorithm. Does a similar algorithm exist for the TSP? If yes, explain briefly how it works. If not, why is this seemingly similar problem so different? (3 points)

Exercise 5: Practice Problems (12 points)

- a) What is the difference between factoring, reverse factoring, and supply chain finance (using the definitions of the lecture)? (3 points)
- b) Comment on the following statement: "Supply chain finance is a win-win-win solution and so suppliers typically join programs quite quickly; getting them onboard is a minor problem." (3 points)

Consider the HP case as we discussed it in class. This case takes place about 30 years ago. Assume there is a firm called PH which faces 100% the same problem but in 2020.

- c) What is the key problem faced by PH (2 points)?
- d) What are the root causes of this problem and how can PH use technology to address this problem? (4 points)